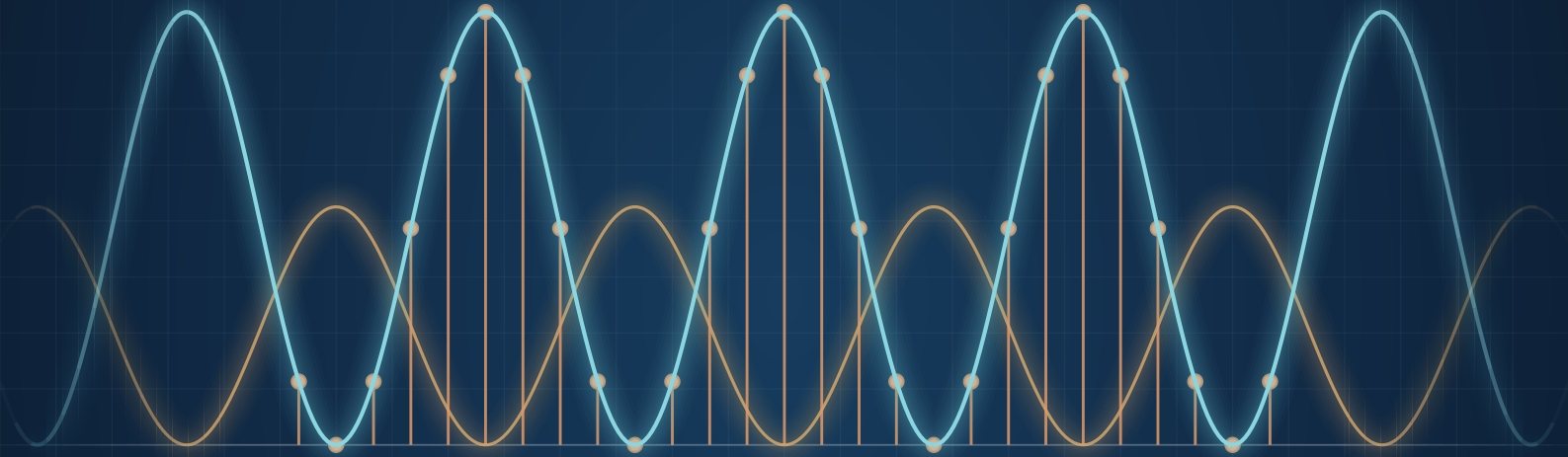




QUANTUM COMPUTING

Quantum Computing

Formula and Notation Reference



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QUANTUM COMPUTING

Formula and Notation Reference

The conventions used throughout the course, every formula it establishes, and every symbol it defines. Nothing here is derived; the derivations are in the lecture notes.

Contains	Edition	Companion
Conventions, formulas, notation	1.0	Lecture notes, Chapters 1 to 6

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How to read it

Part 1 states the four conventions this course fixes and the two expressions everything else is built on. Part 2 is the summary of formulas, in the order the course establishes them. Part 3 defines every symbol. Nothing here is derived: where a result needs an argument, the argument is in the lecture notes chapter named beside it.

This course is an adaptation. Its syllabus and the sequence of topics it teaches derive from **Quantum Computing Lectures** by Aleksandr Krasnok, github.com/AlexKrasnok/quantum-computing-lectures, used under [CC BY 4.0](#). Every scene, figure, laboratory and question here is written for this artifact.

PART

1

Conventions

These hold everywhere in the course, without local variation. The first two are the ones that fail silently. A wrong qubit order does not make a result approximately wrong — it names a different state, and every number after it is quietly wrong. A phase dropped in the wrong place removes the only mechanism any algorithm here has.

CONVENTION	STATEMENT
Qubit order	A register of n qubits is written $ q_{n-1} \dots q_1 q_0\rangle$, and entry x of its column of amplitudes is the amplitude of $ x\rangle$ with x read as a binary number. Circuit drawings put q_0 at the top. Test it before trusting any two-qubit result: prepare $ 10\rangle$, apply X to the qubit you mean, and print all four amplitudes.
Global and relative phase	$e^{i\gamma} \psi\rangle$ is the same state as $ \psi\rangle$, and the phase may always be dropped. $\alpha 0\rangle + e^{i\varphi}\beta 1\rangle$ is not the same state as $\alpha 0\rangle + \beta 1\rangle$, and that phase may never be dropped. Interference is entirely a statement about the second kind.
The inner product	$\langle u v\rangle = \sum_k u_k^* v_k$: the conjugate sits on the first argument and on nothing else. In NumPy that is <code>np.vdot(u, v)</code> and never <code>np.dot</code> . Matrix products act on a ket from right to left, so the gate written last in a product is the one applied first.
Units and logarithms	$\hbar = 1$, so a Hamiltonian is measured in angular frequency and evolution is $U(t) = e^{-iHt}$. Where a physical energy is meant, the units are written out. <code>log</code> without a base means base two, so every entropy and every bit count is in bits.
States and operators	$ \psi\rangle$ is a pure state and ρ a density operator, both normalised: $\langle\psi \psi\rangle = 1$ and $\text{Tr } \rho = 1$. $A^\dagger$ is the conjugate transpose, I the identity, and X, Y, Z the three Pauli operators, also written $\sigma_x, \sigma_y, \sigma_z$.
Exact against sampled	p is an exact Born probability and $\hat{p} = K/N$ is an estimate from N shots of the same circuit. The two differ by about $1/(2\sqrt{N})$ at worst, so a quoted frequency always carries its shot count.
Oracles and cost	$U_f x\rangle y\rangle = x\rangle y \oplus f(x)\rangle$ is one query. A query count is not a runtime, and a resource claim names five things: the task, the input model, the accuracy, the hardware model, and the classical baseline.

THE TWO EXPRESSIONS THE WHOLE COURSE RESTS ON

$$p(n) = |\langle n | \psi \rangle|^2$$

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t) = e^{-iHt}$$

The first is the Born rule, and it is the only place a state becomes a number an instrument can print. The second is closed-system evolution, and every gate in the course is this expression for some H and some t . Measurement and evolution are the two things the postulates license; everything in Part 2 is one of them carrying more structure.

PART

2

Summary of formulas

Every formula the course establishes, in the order the course establishes it. Nothing here is derived and nothing here is new: each entry names the section that develops it, and the argument is in that section. Read this page to check a result you already understand, never to meet one for the first time.

Four conventions are in force throughout and are not repeated under each entry. A register of n qubits is written $|q_{n-1} \dots q_1 q_0\rangle$, and entry x of its column of amplitudes is the amplitude of $|x\rangle$ with x read as a binary number. A phase on the whole state is not physical; a phase between two terms is. The inner product conjugates its first argument. And $\hbar = 1$, so a Hamiltonian is in angular frequency, evolution is $U(t) = e^{-iHt}$, and a logarithm of a probability is taken base two.

A.1 Chapter 1 • The mathematics of quantum states

A QUBIT STATE

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}, \quad |\alpha|^2 + |\beta|^2 = 1$$

The object every other formula acts on. Chapter 1, section 1.1.

THE INNER PRODUCT

$$\langle a| = |a\rangle^\dagger = [a_1^* \quad \dots \quad a_n^*]$$

$$\langle a|b\rangle = \sum_k a_k^* b_k$$

The conjugate sits on the first argument and on nothing else, so $\langle b|a\rangle = \langle a|b\rangle^*$. Chapter 1, section 1.1.

A COEFFICIENT IS AN INNER PRODUCT

$$|v\rangle = \sum_i v_i |e_i\rangle \quad \implies \quad v_j = \langle e_j|v\rangle$$

True only for an orthonormal basis, and it is what an orthonormal basis is chosen for. Chapter 1, section 1.1.

CAUCHY-SCHWARZ, AND THE TWO ENDS OF IT

$$|\langle a|b\rangle| \leq \|a\| \|b\|, \quad 0 \leq |\langle a|b\rangle|^2 \leq 1 \text{ for normalised states}$$

Zero means orthogonal and one measurement separates the two with certainty; one means the same state up to a global phase. Chapter 1, section 1.1.

MODULUS AND PHASE

$$z = x + iy = r e^{i\varphi}, \quad zz^* = x^2 + y^2 = |z|^2$$

Every probability in the course is a $|z|^2$ and every interference effect is a difference of two φ . Chapter 1, section 1.2.

A GLOBAL PHASE IS NOT PHYSICAL, A RELATIVE PHASE IS

$$e^{i\gamma}|\psi\rangle \equiv |\psi\rangle$$

$$\alpha|0\rangle + e^{i\varphi}\beta|1\rangle \not\equiv \alpha|0\rangle + \beta|1\rangle$$

The second line is the mechanism every algorithm in the course runs on. Chapter 1, section 1.2.

THE OUTER PRODUCT AND THE PROJECTOR

$$|a\rangle\langle b| = |a\rangle|b\rangle^\dagger, \quad (|a\rangle\langle b|)_{jk} = a_j b_k^*$$

$$P_u = |u\rangle\langle u|, \quad P_u^\dagger = P_u, \quad P_u^2 = P_u$$

A bra eats the ket to its right first, so $(|a\rangle\langle b|)|v\rangle = \langle b|v\rangle|a\rangle$. Chapter 1, section 1.3.

THE RESOLUTION OF THE IDENTITY

$$\sum_k |e_k\rangle\langle e_k| = I$$

Used as a move rather than as a fact: inserting it turns a statement about a vector into a statement about its coefficients. Chapter 1, section 1.3.

GRAM-SCHMIDT

$$u_j = v_j - \sum_{i<j} \langle e_i|v_j\rangle e_i, \quad e_j = \frac{u_j}{\|u_j\|}$$

The modified recursion subtracts against the partly built u_j rather than against v_j , and keeps far more digits. Chapter 1, section 1.4.

THE TENSOR PRODUCT

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \otimes \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} a_1 b_1 \\ a_1 b_2 \\ a_2 b_1 \\ a_2 b_2 \end{bmatrix}$$

$$(A \otimes B)_{(i,j)} = A_{ij} B, \quad \dim = 2^n \text{ for } n \text{ qubits}$$

Two qubits are four numbers and not two plus two, and that is where the exponential comes from. Chapter 1, section 1.5.

ADJOINT, HERMITIAN, UNITARY

$$A^\dagger = (A^*)^T, \quad (AB)^\dagger = B^\dagger A^\dagger$$

$$A = A^\dagger \implies \text{real eigenvalues}, \quad U^\dagger U = I \iff U^{-1} = U^\dagger$$

A Hermitian operator is a measurable quantity and a unitary one is a reversible motion. Chapter 1, section 1.6.

THE EXPONENTIAL THAT JOINS THEM

$$G = G^\dagger, \theta \in \mathbb{R} \implies U(\theta) = e^{-i\theta G} \text{ is unitary}$$

$$e^{-i\theta \sigma/2} = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) \sigma$$

The second line holds for any operator whose square is the identity, which is every Pauli and every $\mathbf{n} \cdot \boldsymbol{\sigma}$. Chapter 1, section 1.6.

THE SPECTRAL DECOMPOSITION, AND A FUNCTION OF AN OPERATOR

$$A = \sum_k \lambda_k P_k, \quad P_j P_k = \delta_{jk} P_k, \quad \sum_k P_k = I$$

$$f(A) = \sum_k f(\lambda_k) P_k, \quad e^{-iAt} = \sum_k e^{-i\lambda_k t} P_k$$

The second line is what makes an evolution operator computable at all. Chapter 1, section 1.7.

FUNCTIONS AS VECTORS, PARSEVAL AND TRUNCATION

$$\langle f|g \rangle = \int_a^b f^*(x)g(x) dx, \quad |f\rangle = \sum_{n=1}^{\infty} c_n |u_n\rangle, \quad c_n = \langle u_n|f \rangle$$

$$\|f\|^2 = \sum_{n=1}^{\infty} |c_n|^2, \quad \left\| f - \sum_{n=1}^N c_n u_n \right\|^2 = \sum_{n>N} |c_n|^2$$

The coefficient tail is the exact squared norm error of a truncated complete orthonormal expansion. Chapter 1, section 1.9.

A.2 Chapter 2 · States, measurement and dynamics

THE BORN RULE

$$p(n) = |\langle n|\psi\rangle|^2$$

The one postulate that connects a state to a number an instrument prints. Chapter 2, section 2.1.

PASSIVE COORDINATES AGAINST AN ACTIVE TRANSFORMATION

$$|\psi'\rangle = V^\dagger|\psi\rangle, \quad A' = V^\dagger A V, \quad \langle\psi|A|\psi\rangle = \langle\psi'|A'|\psi'\rangle$$

$$|\psi\rangle \mapsto U|\psi\rangle \quad \text{with } A \text{ fixed}$$

Passive coordinates preserve every prediction; an active operation generally changes them. Choosing new measurement projectors is a third operation. Chapter 2, section 2.1.

A PROJECTIVE MEASUREMENT, AND THE STATE IT LEAVES

$$p(a) = \langle\psi|P_a|\psi\rangle, \quad P_j P_k = \delta_{jk} P_k, \quad \sum_a P_a = I$$

$$|\psi_a\rangle = \frac{P_a|\psi\rangle}{\sqrt{p(a)}}$$

Orthogonality makes the outcomes exclusive and completeness makes them add to one. Chapter 2, section 2.2.

A POVM AND ITS MEASUREMENT INSTRUMENT

$$E_m = \sum_\alpha M_{m\alpha}^\dagger M_{m\alpha}, \quad p(m) = \text{Tr}(\rho E_m), \quad \sum_m E_m = I$$

$$\rho_m = \frac{\sum_\alpha M_{m\alpha} \rho M_{m\alpha}^\dagger}{p(m)}$$

The effects fix the probabilities; the instrument also fixes the conditional state. The projective case is $M_m = P_m$. Chapter 2, section 2.2.

EXPECTATION AND VARIANCE

$$\langle A \rangle = \langle\psi|A|\psi\rangle = \sum_a a p(a)$$

$$\text{Var}(A) = \langle A^2 \rangle - \langle A \rangle^2, \quad \Delta A = \sqrt{\text{Var}(A)}$$

For any Pauli $A^2 = I$, so $\text{Var}(A) = 1 - \langle A \rangle^2$ and no second sandwich is needed. Chapter 2, section 2.3.

THE ROBERTSON RELATION

$$\Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|, \quad [A, B] = AB - BA$$

The bound depends on the state, so a vanishing right-hand side says nothing rather than saying both spreads are zero. Chapter 2, section 2.4.

OBSERVABLE

+1 EIGENSTATE

−1 EIGENSTATE

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$|+\rangle = \frac{1}{\sqrt{2}}(1, 1)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(1, -1)$$

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$|+i\rangle = \frac{1}{\sqrt{2}}(1, i)$$

$$|-i\rangle = \frac{1}{\sqrt{2}}(1, -i)$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$|0\rangle = (1, 0)$$

$$|1\rangle = (0, 1)$$

THE PAULI ALGEBRA

$$\sigma_i \sigma_j = \delta_{ij} I + i \sum_k \varepsilon_{ijk} \sigma_k$$

$$[\sigma_i, \sigma_j] = 2i \sum_k \varepsilon_{ijk} \sigma_k, \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij} I$$

Cyclic order $X \rightarrow Y \rightarrow Z \rightarrow X$ carries the plus sign. Two different Pauli operators anticommute. Chapter 2, section 2.5.

MEASURING ALONG A DIRECTION

$$\mathbf{n} \cdot \boldsymbol{\sigma}, \quad P_{\pm} = \frac{1}{2} (I \pm \mathbf{n} \cdot \boldsymbol{\sigma}), \quad r_a = \langle \sigma_a \rangle$$

$$p(\pm) = \frac{1}{2} (1 \pm \mathbf{n} \cdot \mathbf{r}), \quad p(+) = \cos^2 \frac{\alpha}{2}$$

α is the angle between two vectors, which is twice the angle between the two states. Chapter 2, section 2.5.

COORDINATE OPERATORS AND A FREE PARTICLE

$$(\hat{x}\psi)(x) = x\psi(x), \quad (\hat{p}\psi)(x) = -i\hbar \frac{d\psi}{dx}$$

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}, \quad p = \hbar k, \quad E = \frac{\hbar^2 k^2}{2m}$$

A plane wave is a generalised eigenfunction; a physical free-particle state is a normalisable wave packet. Chapter 2, section 2.6.

THE INFINITE SQUARE WELL

$$\phi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), \quad n = 1, 2, \dots$$

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2ma^2}$$

The boundary conditions select discrete wave numbers. There is no $n = 0$ state, and $E_2 = 4E_1$. Chapter 2, section 2.6.

CLOSED-SYSTEM EVOLUTION

$$i \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle$$

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad U(t) = e^{-iHt}$$

Replacing H by $H + cI$ multiplies U by a global phase, so only energy differences are physical. Chapter 2, section 2.6.

A DRIVEN QUBIT

$$H = \frac{1}{2} (\Omega_x X + \Omega_y Y + \Delta Z) = \frac{\Omega}{2} \mathbf{n} \cdot \boldsymbol{\sigma}, \quad \Omega = \sqrt{\Omega_x^2 + \Omega_y^2 + \Delta^2}$$

$$U(t) = \cos\left(\frac{\Omega t}{2}\right) I - i \sin\left(\frac{\Omega t}{2}\right) \mathbf{n} \cdot \boldsymbol{\sigma}, \quad p_{\max} = \left(\frac{\Omega_x}{\Omega}\right)^2$$

Pulse area sets the angle, drive phase sets the axis in the equator, and detuning tilts that axis towards z . Chapter 2, section 2.6.

WHAT A FINITE RUN IS WORTH

$$K \sim \text{Binomial}(N, p), \quad \text{SE}\left(\frac{K}{N}\right) = \sqrt{\frac{p(1-p)}{N}} \leq \frac{1}{2\sqrt{N}}$$

One more decimal place costs a hundred times the shots. This is counting and not physics, so no hardware improvement changes it. Chapter 2, section 2.7.

A.3 Chapter 3 • Mixed states and entanglement

THE DENSITY OPERATOR

$$\rho_\psi = |\psi\rangle\langle\psi|$$

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1$$

It does not remember which preparation made it: two different mixtures with the same ρ are the same state. Chapter 3, section 3.1.

WHICH MATRICES ARE STATES, AND EVERY PREDICTION FROM ONE TRACE

$$\rho = \rho^\dagger, \quad \rho \succeq 0, \quad \text{Tr } \rho = 1$$

$$\langle A \rangle = \text{Tr}(\rho A), \quad p(m) = \text{Tr}(\rho E_m)$$

For one qubit positivity is the single inequality $|\rho_{01}|^2 \leq \rho_{00}\rho_{11}$. Chapter 3, section 3.1.

PURITY

$$\gamma = \text{Tr}(\rho^2) = \sum_k \lambda_k^2, \quad \frac{1}{d} \leq \gamma \leq 1$$

It says how mixed and never which state: one number where a qubit state needs three. Chapter 3, section 3.2.

A QUBIT STATE AS A VECTOR

$$\rho = \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}), \quad r_a = \text{Tr}(\rho \sigma_a), \quad |\mathbf{r}| \leq 1$$

$$\lambda_\pm = \frac{1}{2}(1 \pm |\mathbf{r}|), \quad \text{Tr } \rho^2 = \frac{1}{2}(1 + |\mathbf{r}|^2)$$

From a state vector (a, b) the y component reads $r_y = 2 \text{Im}(a^*b)$; from ρ it reads $r_y = -2 \text{Im } \rho_{01}$. The two signs are not a contradiction, and both give $r_y = +1$ on $|+i\rangle$. Chapter 3, section 3.2.

A CHANNEL IN KRAUS FORM

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I$$

Every channel arises this way, and two different sets of Kraus operators can describe one channel. Chapter 3, section 3.3.

AMPLITUDE DAMPING

$$K_0 = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{bmatrix}, \quad K_1 = \begin{bmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{bmatrix}$$

$$\rho_{11} \mapsto (1-\gamma)\rho_{11}, \quad \rho_{01} \mapsto \sqrt{1-\gamma}\rho_{01}$$

At $\gamma = 1$ every state becomes $|0\rangle$, which is why this is the model of relaxation. Chapter 3, section 3.3.

THE PHASE-FLIP CHANNEL

$$\mathcal{E}_Z(\rho) = (1-p)\rho + pZ\rho Z$$

$$\rho_{00}, \rho_{11} \text{ unchanged}, \quad \rho_{01} \mapsto (1-2p)\rho_{01}$$

The damage peaks at $p = \frac{1}{2}$ and vanishes at $p = 1$, where the channel is the gate Z . Chapter 3, section 3.3.

THE TWO DECAY TIMES

$$\rho_{11}(t) = \rho_{11}^{\text{eq}} + [\rho_{11}(0) - \rho_{11}^{\text{eq}}] e^{-t/T_1}, \quad \rho_{01}(t) = e^{-t/T_2} \rho_{01}(0)$$

$$\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi} \quad \implies \quad T_2 \leq 2T_1$$

Populations decay on T_1 and the coherence on T_2 , and the inequality is a theorem rather than a measurement. Chapter 3, section 3.4.

THE ORDERING THIS COURSE FIXES

$$\begin{bmatrix} a \\ b \end{bmatrix} \otimes \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} ac \\ ad \\ bc \\ bd \end{bmatrix}, \quad |q_1 q_0\rangle$$

Read under the other convention this names a different state, and every number downstream is quietly wrong. Chapter 3, section 3.5.

THE PARTIAL TRACE

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \sum_j (I_A \otimes \langle j|) \rho_{AB} (I_A \otimes |j\rangle)$$

$$\text{Tr}(\rho_A A) = \text{Tr}[\rho_{AB} (A \otimes I_B)] \quad \text{for every } A$$

The second line is the property that makes it the right operation: it is the state that answers every question about A alone. Chapter 3, section 3.5.

SEPARABILITY, AND THE SCHMIDT FORM

$$|\psi\rangle \text{ is a product } \iff c_0c_3 - c_1c_2 = 0$$

$$|\psi\rangle_{AB} = \sum_{k=1}^r \sqrt{\lambda_k} |u_k\rangle_A |v_k\rangle_B, \quad \lambda_k > 0, \quad \sum_k \lambda_k = 1$$

The Schmidt rank r is one exactly when the pair is a product. Chapter 3, section 3.6.

ENTANGLEMENT ENTROPY

$$S(\rho) = -\text{Tr}(\rho \log_2 \rho) = -\sum_k \lambda_k \log_2 \lambda_k, \quad S(\rho_A) = S(\rho_B)$$

Zero for a product and one bit for a Bell state. Chapter 3, section 3.6.

THE BELL STATES

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \quad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

$$\text{for } |\Phi^+\rangle : \quad \langle X \otimes X \rangle = 1, \quad \langle Y \otimes Y \rangle = -1, \quad \langle Z \otimes Z \rangle = 1$$

All four share the same reduced state $I/2$ and differ only in their joint correlations. Chapter 3, section 3.7.

CHSH, AND THE TWO CEILINGS

$$S = \langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle$$

$$|S| \leq 2 \text{ for pre-existing values,} \quad |S| \leq 2\sqrt{2} \text{ for quantum states}$$

Both bounds are theorems. A measured value above two is what rules out the first model. Chapter 3, section 3.7.

A.4 Chapter 4 • The Bloch sphere and quantum gates

A PURE QUBIT STATE, AND THE POINT IT NAMES

$$|\psi(\theta, \varphi)\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi < 2\pi$$

$$\mathbf{r} = (\sin \theta \cos \varphi, \sin \theta \sin \varphi, \cos \theta), \quad |\mathbf{r}| = 1$$

At a pole the azimuth means nothing, because $\sin 0 = 0$ leaves nothing for the phase to multiply. Chapter 4, section 4.1.

THE OVERLAP OF TWO PURE STATES

$$|\langle \chi | \psi \rangle|^2 = \frac{1 + \mathbf{r} \cdot \mathbf{s}}{2} = \cos^2 \frac{\Theta}{2}$$

Orthogonal states sit at opposite points; points at a right angle overlap with probability one half. Chapter 4, section 4.1.

A FULL TURN IS NOT THE IDENTITY

$$R_{\mathbf{n}}(2\pi) = -I, \quad R_{\mathbf{n}}(4\pi) = +I$$

U and $-U$ give the same rotation of the sphere, so the map from gates to rotations is two to one. Chapter 4, section 4.2.

THE ROTATION OPERATOR

$$R_{\mathbf{n}}(\alpha) = e^{-i\alpha \mathbf{n} \cdot \boldsymbol{\sigma}/2} = \cos \frac{\alpha}{2} I - i \sin \frac{\alpha}{2} \mathbf{n} \cdot \boldsymbol{\sigma}$$

$$\text{every } 2 \times 2 \text{ unitary} = e^{i\gamma} R_{\mathbf{n}}(\alpha) \text{ for some } \gamma, \mathbf{n}, \alpha$$

Rotation is not one family of one-qubit gates among many; it is all of them. Chapter 4, section 4.3.

THE PAULI GATES AS ROTATIONS

$$R_x(\pi) = -iX, \quad R_y(\pi) = -iY, \quad R_z(\pi) = -iZ$$

$$X : |0\rangle \leftrightarrow |1\rangle, \quad |+\rangle \text{ and } |-\rangle \text{ fixed}$$

$$Z : |+\rangle \leftrightarrow |-\rangle, \quad |0\rangle \text{ and } |1\rangle \text{ fixed}$$

Which gate looks like a flip depends entirely on the basis the state is written in. Chapter 4, section 4.3.

THE HADAMARD

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{X+Z}{\sqrt{2}}, \quad H^2 = I$$

$$H X H = Z, \quad H Z H = X, \quad H Y H = -Y$$

Measuring X is applying H and then measuring Z , which is the only reading a machine offers. Chapter 4, section 4.3.

THE PHASE FAMILY

$$P(\varphi) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix} = e^{i\varphi/2} R_z(\varphi)$$

$$S = P\left(\frac{\pi}{2}\right), \quad T = P\left(\frac{\pi}{4}\right), \quad S^2 = Z, \quad T^2 = S$$

S is a Clifford gate and cheap to correct for; T is not, and that difference sets the cost of fault tolerance. Chapter 4, section 4.3.

THE ORDER A CIRCUIT MULTIPLIES IN

$$U_1 \text{ first, } U_2 \text{ next, } U_3 \text{ last} \implies U = U_3 U_2 U_1$$

$$(A \otimes I)(I \otimes B) = A \otimes B = (I \otimes B)(A \otimes I)$$

Gates on different qubits always commute; gates sharing a qubit usually do not. Chapter 4, section 4.4.

THE EULER DECOMPOSITION, AND THE GATE WITH THREE DIALS

$$U = e^{i\alpha} R_z(\phi) R_y(\theta) R_z(\lambda)$$

$$U(\theta, \phi, \lambda) = \begin{bmatrix} \cos \frac{\theta}{2} & -e^{i\lambda} \sin \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} & e^{i(\phi+\lambda)} \cos \frac{\theta}{2} \end{bmatrix}$$

$H = U(\frac{\pi}{2}, 0, \pi)$, $X = U(\pi, 0, \pi)$ and $P(\varphi) = U(0, \varphi, 0)$. Chapter 4, section 4.4.

REVERSIBLE EMBEDDINGS AND THE ORACLE PATTERN

$$(a, b) \mapsto (a, a \oplus b) \quad \text{is exactly CNOT}$$

$$|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle \quad \text{for any } f$$

AND needs a third wire, $(a, b, c) \mapsto (a, b, c \oplus ab)$, which is the Toffoli gate. Chapter 4, section 4.5.

A REVERSIBLE HALF ADDER

$$(a, b, 0, 0) \mapsto (a, b, a \oplus b, ab)$$

Two CNOTs write the sum and one Toffoli writes the carry. Preserving the inputs makes the map invertible. Chapter 4, section 4.5.

COMPUTE, COPY OUT, UNCOMPUTE

$$V_f^\dagger \text{ (copy) } V_f : |x\rangle|0\rangle|y\rangle \mapsto |x\rangle|0\rangle|y \oplus f(x)\rangle$$

An ancilla that leaves in a state depending on x stays entangled with the register, and the interference at the end does not happen. Chapter 4, section 4.5.

A ONE-QUBIT GATE ON A NAMED QUBIT

$$\begin{aligned} \text{on } q_0 : I \otimes A, \quad \text{on } q_1 : A \otimes I \\ (I \otimes X) |10\rangle = |11\rangle, \quad (X \otimes I) |10\rangle = |00\rangle \end{aligned}$$

Both are valid gates and they do different things. Nothing in the mathematics protects a reader who picks the wrong one. Chapter 4, section 4.6.

THE TWO CNOT MATRICES, IN THE BASIS ORDER $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

$$\text{CNOT} : |c\rangle|t\rangle \mapsto |c\rangle|c \oplus t\rangle$$

$$\text{CNOT}_{0 \rightarrow 1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}, \quad \text{CNOT}_{1 \rightarrow 0} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

The subscript reads control to target. Chapter 4, section 4.6.

CZ AND SWAP

$$\text{CZ} = \text{diag}(1, 1, 1, -1) = (H \otimes I) \text{CNOT}_{0 \rightarrow 1} (H \otimes I)$$

$$\text{SWAP} = \text{CNOT}_{0 \rightarrow 1} \text{CNOT}_{1 \rightarrow 0} \text{CNOT}_{0 \rightarrow 1}$$

Three two-qubit gates to move a state one step across a chip, which is what routing costs. Chapter 4, section 4.6.

MAKING A BELL PAIR, AND TWO UNIVERSAL SETS

$$|00\rangle \xrightarrow{I \otimes R_y(\theta)} \cos \frac{\theta}{2} |00\rangle + \sin \frac{\theta}{2} |01\rangle \xrightarrow{\text{CNOT}_{0 \rightarrow 1}} \cos \frac{\theta}{2} |00\rangle + \sin \frac{\theta}{2} |11\rangle$$

$$\{\text{every one-qubit gate}\} \cup \{\text{CNOT}\} \text{ exactly,} \quad \{H, S, T, \text{CNOT}\} \text{ to any } \varepsilon > 0$$

A gate is entangling if it entangles some input, not every input. The price of ε is a length growing like a power of $\log(1/\varepsilon)$. Chapter 4, section 4.7.

A.5 Chapter 5 • Circuits and protocols

THE CIRCUIT MODEL, IN ONE LINE

$$|0\rangle^{\otimes n} \xrightarrow{U_d \cdots U_2 U_1} |\psi\rangle \xrightarrow{\text{measure}} x \in \{0, 1\}^n$$

Fixed input, fixed gate set, fixed measurement basis, and one string per run. Chapter 5, section 5.1.

ONE STATE, NAMED FOUR WAYS

$$|q_{n-1} \cdots q_1 q_0\rangle, \quad x = \sum_k 2^k q_k$$

A run that prints 101 read $q_2 = 1, q_1 = 0, q_0 = 1$, which is entry 5 of the state vector. Chapter 5, section 5.1.

DEPTH, GATE COUNT AND SIMULATION COST

depth = layers of gates that must run in order, count = gates

$$\text{bytes} = 16 \cdot 2^n, \quad n = 30 \Rightarrow 17 \text{ GB}, \quad n = 50 \Rightarrow 18 \text{ PB}$$

A coherence time is spent against depth and an error budget against the two-qubit count. The size of the vector is an upper bound on the difficulty of simulation and never a lower one. Chapter 5, sections 5.1 and 5.2.

THE ERROR BAR ON A PROBABILITY READ FROM A FINITE RUN

$$K \sim \text{Binomial}(N, p), \quad \text{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{N}}$$

An estimate near one half to within ± 0.005 needs ten thousand shots. Chapter 5, section 5.2.

DEFERRED MEASUREMENT

measure q_0 , then X^m on $q_1 \equiv \text{CNOT}_{0 \rightarrow 1}$, then measure q_0

The rule is about a measured wire used as a control and about nothing else. Chapter 5, section 5.2.

TWO EXACT REWRITES A COMPILER USES

$$H = e^{i\pi/2} R_z\left(\frac{\pi}{2}\right) R_x\left(\frac{\pi}{2}\right) R_z\left(\frac{\pi}{2}\right)$$

$$\text{CNOT}_{0 \rightarrow 1} = (H \text{ on } q_1) \text{ CZ } (H \text{ on } q_1)$$

What is lost is only length; the two-qubit count does not move. The phase in the first line becomes observable the moment the gate is controlled. Chapter 5, section 5.3.

THE FAULT-TOLERANCE THRESHOLD CONDITION

$$p < p_{\text{th}}$$

For a stated code, noise model and control system, increasing the code distance can then reduce the logical error rate. The threshold is not one universal number. Chapter 5, section 5.3.

TURNING A PHASE INTO COUNTS

$$\begin{aligned} |0\rangle &\xrightarrow{H} \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \xrightarrow{P(\varphi)} \frac{1}{\sqrt{2}} (|0\rangle + e^{i\varphi}|1\rangle) \\ &\xrightarrow{H} \frac{1}{2} (1 + e^{i\varphi}) |0\rangle + \frac{1}{2} (1 - e^{i\varphi}) |1\rangle, \quad p(0) = \cos^2 \frac{\varphi}{2} \end{aligned}$$

Measuring in the middle learns nothing; the second Hadamard is what makes the phase readable. Chapter 5, section 5.4.

TELEPORTATION, AS ONE IDENTITY

$$|\Psi_2\rangle = \frac{1}{2} \sum_{m_1, m_0 \in \{0,1\}} |m_1 m_0\rangle_{10} \otimes X^{m_1} Z^{m_0} |\psi\rangle_2$$

Exact, and independent of α and β . Bob applies Z^{m_0} and then X^{m_1} to undo it. Chapter 5, section 5.5.

WHAT BOB HOLDS BEFORE THE BITS ARRIVE

$$\rho_B = \frac{1}{4} \sum_{m_1, m_0} X^{m_1} Z^{m_0} \rho Z^{m_0} X^{m_1} = \frac{I}{2}$$

No ρ on the right-hand side, which is why the protocol sends nothing ahead of the two classical bits. Chapter 5, section 5.5.

PHASE KICKBACK

$$\begin{aligned} X|-\rangle &= -|-\rangle \\ U_f |x\rangle|-\rangle &= (-1)^{f(x)} |x\rangle|-\rangle \end{aligned}$$

The target comes out unchanged, so the answer has landed on the first register as a sign. Chapter 5, section 5.6, and again in chapter 6.

THE PLANE GROVER SEARCH LIVES IN

$$|G\rangle = \frac{1}{\sqrt{M}} \sum_{f(x)=1} |x\rangle, \quad |B\rangle = \frac{1}{\sqrt{N-M}} \sum_{f(x)=0} |x\rangle$$

$$|s\rangle = \sin \theta |G\rangle + \cos \theta |B\rangle, \quad \sin \theta = \sqrt{\frac{M}{N}}$$

A real two-dimensional plane inside a complex space of 2^n dimensions. It is not a Bloch sphere. Chapter 5, section 5.6.

THE GROVER SUCCESS PROBABILITY, AND WHERE IT PEAKS

$$P_{\text{good}}(r) = \sin^2((2r+1)\theta)$$

$$r_* = \frac{\pi}{4\theta} - \frac{1}{2}, \quad r_* \approx \frac{\pi}{4} \sqrt{\frac{N}{M}} \text{ when } M \ll N$$

Each iteration adds a fixed angle 2θ , and a right angle has to be reached. Iterating past r_* makes the answer less likely. Chapter 5, section 5.6.

✔ WHAT A RESOURCE CLAIM MUST NAME

The task, the input model, the accuracy, the hardware model, and the classical baseline it is measured against. A claim missing one of the five is not yet a claim, and a query count is not a runtime. Chapter 5, section 5.7.

A.6 Chapter 6 • Quantum algorithms

THE ORACLE, AND WHAT ONE QUERY MEANS

$$U_f |x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$$

Building U_f costs gates and those gates are not counted. A query separation is a theorem about one column of the table. Chapter 6, section 6.1.

KICKBACK WITH AN ARBITRARY UNITARY

$$U |u\rangle = e^{2\pi i \varphi} |u\rangle$$

$$cU \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|u\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i \varphi}|1\rangle)|u\rangle$$

On a superposition of eigenstates the procedure samples one eigenphase with probability $|c_k|^2$; it does not list the spectrum. Chapter 6, section 6.2.

WHAT THE LAST LAYER OF HADAMARDS IS FOR

$$\langle 0^n | H^{\otimes n} \left(\frac{1}{2^{n/2}} \sum_x (-1)^{f(x)} |x\rangle \right) = \frac{1}{2^n} \sum_x (-1)^{f(x)}$$

A superposition that never interferes has bought nothing. The readout returns n bits, never 2^n numbers. Chapter 6, section 6.2.

DEUTSCH, AND DEUTSCH-JOZSA

$$\frac{1}{\sqrt{2}} \left((-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle \right) \xrightarrow{H} \pm |f(0) \oplus f(1)\rangle$$

$$\text{amplitude of } |0^n\rangle = \frac{1}{2^n} \sum_x (-1)^{f(x)} = \begin{cases} \pm 1 & \text{constant} \\ 0 & \text{balanced} \end{cases}$$

One query against $2^{n-1} + 1$ exact classical queries, on a promised input. Chapter 6, section 6.3.

THE QUANTUM FOURIER TRANSFORM, AND ITS CIRCUIT

$$F_Q |x\rangle = \frac{1}{\sqrt{Q}} \sum_{k=0}^{Q-1} e^{2\pi i xk/Q} |k\rangle, \quad Q = 2^n$$

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{bmatrix}, \quad n + \frac{1}{2}n(n-1) = \frac{1}{2}n(n+1) \text{ gates}$$

It is a change of basis and returns one index, never a spectrum. F_2 is the Hadamard. Chapter 6, section 6.4.

PHASE ESTIMATION

$$\frac{1}{\sqrt{2^t}} \sum_{k=0}^{2^t-1} e^{2\pi i k \varphi} |k\rangle \xrightarrow{F^\dagger} y, \quad \hat{\varphi} = \frac{y}{2^t}$$

$$P(y) = \frac{1}{2^{2t}} \left| \frac{\sin(\pi 2^t \delta)}{\sin(\pi \delta)} \right|^2, \quad \delta = \varphi - \frac{y}{2^t}$$

Costs $2^t - 1$ applications of U against $\frac{1}{2}t(t+1)$ gates in the inverse transform. Chapter 6, section 6.5.

THE TWO FLOORS, AND HOW MANY COUNTING QUBITS TO TAKE

$$P(\text{nearest}) \geq \frac{4}{\pi^2} \approx 0.405, \quad P(\text{two nearest}) \geq \frac{8}{\pi^2} \approx 0.811$$

$$t = n + \left\lceil \log_2 \left(2 + \frac{1}{2\varepsilon} \right) \right\rceil$$

Neither floor moves as the register grows: extra qubits buy accuracy and never confidence. Chapter 6, section 6.5.

ORDER FINDING

$$U_a |y\rangle = |a y \bmod N\rangle, \quad U_a |u_s\rangle = e^{2\pi i s/r} |u_s\rangle$$

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle, \quad |u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} |a^k \bmod N\rangle$$

The order sits in the denominators of the eigenphases, and starting the work register in the state one costs a single gate. Chapter 6, section 6.6.

WHY THE READING IS ENOUGH

$$\left| \frac{y}{Q} - \frac{s}{r} \right| \leq \frac{1}{2Q}, \quad Q > N^2 \implies \frac{s}{r} \text{ is the only such fraction}$$

Continued fractions recover s/r from the reading, and one modular exponentiation confirms r . The condition $Q > N^2$ is what doubles the counting register. Chapter 6, section 6.6.

THE FACTORS, FROM THE ORDER

$$\gcd(a^{r/2} - 1, N) \quad \text{and} \quad \gcd(a^{r/2} + 1, N)$$

Needs r even and $a^{r/2} \not\equiv -1 \pmod{N}$; a random a satisfies both at least half the time. Chapter 6, section 6.7.

WHERE THE WORK IS

modular exponentiation $O(L^3)$, transform $O(L^2)$, everything else classical

The arithmetic is the algorithm, and the quantum step exists only to supply the exponent that makes the algebra apply. Chapter 6, section 6.7.

PART

3

Notation

Every symbol the course defines, with the chapter that defines it. A symbol is never reused for a second meaning.

SYMBOL	MEANING
$ \psi\rangle$	A quantum state, written as a column of complex numbers once a basis has been chosen. Normalised: $\langle\psi \psi\rangle = 1$.
$\langle\psi $	The dual of $ \psi\rangle$: the same column transposed and conjugated, so it is a row that eats a ket and returns a number.
$\langle\phi \psi\rangle$	Inner product, conjugating the first argument. It measures overlap: zero when the two states are perfectly distinguishable, unit modulus when they are the same state.
$\ \psi\ $	Length of a state, $\sqrt{\langle\psi \psi\rangle}$. A physical state has length one, which is what makes the Born probabilities sum to one.
$ \phi\rangle\langle\psi $	Outer product: a ket beside a bra is an operator, not a number. It is the one construction that turns states into the things that act on states.
P_k	A projector, $P_k = k\rangle\langle k $ for a basis state. Idempotent, $P_k^2 = P_k$, and Hermitian.
$\sum_k k\rangle\langle k = I$	Resolution of the identity. Inserting it is how a basis expansion is derived rather than guessed.
$\otimes$	Tensor product. Two systems of dimension d_1 and d_2 make one system of dimension $d_1 d_2$, not $d_1 + d_2$: this is where the exponential comes from.
$A^\dagger$	Adjoint: transpose and conjugate. In NumPy, <code>A.conj().T</code> .
$A = A^\dagger$	Hermitian. Its eigenvalues are real, which is why observables are Hermitian: a measurement returns a real number.
$U^\dagger U = I$	Unitary. It preserves inner products and therefore norms, which is why closed-system evolution and every gate is unitary.
$A v\rangle = \lambda v\rangle$	Eigenvalue equation. The eigenvectors are the states the operator leaves in place, up to a scale.
$A = \sum_k \lambda_k P_k$	Spectral decomposition of a Hermitian operator: the eigenvalues, and the projectors onto the eigenspaces they belong to.
$f(A) = \sum_k f(\lambda_k) P_k$	A function of an operator is that function applied to the eigenvalues. This is what an operator exponential means, and it is why e^{-iHt} can be computed at all.
$L^2[a, b]$	The vector space of square-integrable functions on an interval, with inner product $\langle f g\rangle = \int_a^b f^*(x)g(x) dx$.
$\ f\ ^2 = \sum_n c_n ^2$	Parseval identity for a complete orthonormal basis. The omitted coefficient tail is the exact squared error of a truncated expansion.
$ 0\rangle, 1\rangle$	The computational basis of one qubit, the eigenvectors of Z . Every measurement in this course is in this basis unless a scene says otherwise.
$ +\rangle, -\rangle$	The X basis, $ \pm\rangle = (0\rangle \pm 1\rangle)/\sqrt{2}$. A state definite in X is maximally uncertain in Z .
$ +i\rangle, -i\rangle$	The Y basis, $(0\rangle \pm i 1\rangle)/\sqrt{2}$.
$p(k) = \langle k \psi\rangle ^2$	The Born rule. The amplitude is complex and unobservable; its squared modulus is the probability, and that is the only bridge from the mathematics to a laboratory count.

A	An observable: a Hermitian operator whose eigenvalues are the outcomes that can be seen and whose eigenvectors are the states that give them with certainty.
$\langle A \rangle$	Expectation value, $\langle \psi A \psi \rangle$. It is an average over repetitions, never the result of one run.
ΔA	Standard deviation of an observable in a state, $\sqrt{\langle A^2 \rangle - \langle A \rangle^2}$.
$[A, B]$	Commutator, $AB - BA$. It is zero exactly when the two observables share an eigenbasis and can both be definite at once.
X, Y, Z	The Pauli operators. Each is Hermitian and unitary at once, so each is both an observable and a gate.
H	Hamiltonian: the Hermitian operator that generates evolution. With $\hbar = 1$ it is measured in angular frequency.
$U(t) = e^{-iHt}$	The evolution operator of a closed system. Every gate in this course is one of these for some H and some duration.
$\{E_k\}$	A POVM: positive operators summing to the identity. It fixes the outcome probabilities, but not the state left by the measurement.
$\{M_{m\alpha}\}$	A measurement instrument. It realises each POVM effect and fixes the conditional state after an outcome.
$\hat{x}, \hat{p}$	Position and momentum in the coordinate representation: position multiplies a wavefunction and momentum differentiates it.
ϕ_n, E_n	The stationary states and discrete energies of the infinite square well, selected by boundary conditions at the two walls.
N_{shots}	The number of repetitions a circuit is run for. A probability read from N shots carries a standard error of about $\sqrt{p(1-p)/N}$, which is sampling noise and not physical noise.
ρ	Density operator: positive semidefinite, Hermitian, unit trace. It describes a state that is either genuinely mixed or a piece of a larger entangled state.
$\text{Tr } \rho^2$	Purity. Equal to one for a pure state and as low as $1/d$ for the maximally mixed state of dimension d .
$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$	A quantum channel in Kraus form, with $\sum_k K_k^\dagger K_k = I$. This is what a gate becomes once the system is no longer closed.
Tr_B	Partial trace: the operation that answers "what is the state of A alone". Its result is mixed exactly when A and B are entangled.
λ_i	Schmidt coefficients of a bipartite pure state. One non-zero coefficient means the state is a product; more than one means it is entangled.
$S(\rho)$	Von Neumann entropy, $-\text{Tr } \rho \log_2 \rho$, in bits. For a pure state it is zero, and for the reduced state of an entangled pair it is how much entanglement the pair carries.
$ \Phi^\pm\rangle, \Psi^\pm\rangle$	The four Bell states: an orthonormal basis of two qubits in which every state is maximally entangled.

T_1, T_2	Relaxation and dephasing times. T_1 is how long a population survives, T_2 how long a relative phase does, and $T_2 \leq 2T_1$ always.
$\mathbf{r}$	Bloch vector, $r_a = \text{Tr}(\rho \sigma_a)$ for $a = x, y, z$. Its length is one for a pure state and less for a mixed one, so the sphere is really a ball.
θ, φ	Polar and azimuthal angles of a pure qubit state, $\cos(\theta/2) 0\rangle + e^{i\varphi} \sin(\theta/2) 1\rangle$. The half angle is not a typographic accident: it is the double cover.
$R_a(\theta)$	Rotation of the Bloch vector by θ about axis a , $e^{-i\theta\sigma_a/2}$. Every single-qubit gate is one of these up to a global phase.
H	The Hadamard gate, which exchanges the Z and X bases. Written the same way as a Hamiltonian, and told apart by where it stands: a gate acts in a circuit, a Hamiltonian sits in an exponential.
S, T	The quarter- and eighth-turn phase gates about Z , $S = \text{diag}(1, i)$ and $T = \text{diag}(1, e^{i\pi/4})$.
CNOT	The controlled-NOT gate. Together with the single-qubit gates it is universal, and on its own it is what turns a product state into an entangled one.
d	Circuit depth: the number of layers of gates that must run one after another. It is the quantity a coherence time is spent against, and it is not the gate count.
ISA	The instruction set a particular machine actually runs. A circuit written in ideal gates is transpiled into it, and the transpiled depth is the one that matters.
F	Fidelity between a prepared state and its target, $F = \langle \psi \rho \psi \rangle$ for a pure target. A protocol is claimed to work only against a fidelity that beats the best classical strategy.
k_{opt}	The number of Grover iterations that maximises the success probability, about $\frac{\pi}{4} \sqrt{N/M}$. Running past it makes the answer less likely, not more.
U_f	A reversible embedding of a classical function, $U_f x\rangle y\rangle = x\rangle y \oplus f(x)\rangle$. Counting queries to it is not the same as counting the gates it costs to build.
phase kickback	The mechanism behind every algorithm in this chapter: an oracle written to change a target register instead writes a phase onto the control register, because the target was prepared in an eigenstate of the operation.
QFT_N	The quantum Fourier transform on $N = 2^n$ amplitudes. It costs $O(n^2)$ gates, and it does not hand back the spectrum: the amplitudes it produces still have to be measured.
φ	The phase estimated by quantum phase estimation, defined by $U u\rangle = e^{2\pi i \varphi} u\rangle$. The number of counting qubits fixes both the precision and the success probability.
r	The order of a modulo N : the smallest $r > 0$ with $a^r \equiv 1 \pmod{N}$. Finding it is the only quantum step in factoring; everything around it is classical.

**Quantum Computing · Formula and Notation Reference**

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